Parent Guide — senior / module-07-eleven-multiplication

IKS Curriculum

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Parent Guide — ×11 Multiplication — Ekādhikena Pūrveṇa (Senior)

What is your child learning?

A mental-arithmetic method for multiplying any number by 11 instantly: split the digits, add adjacent pairs, place sums in between. Tirthaji's *Vedic Mathematics* (1965) labelled this an application of the *Ekādhikena Pūrveṇa* sūtra. Honest history is taught alongside: this is a 20th-century mental-arithmetic system, not strictly ancient mathematics.

Why this matters

This module is part of an Indian Knowledge Systems curriculum that introduces classical Indian frameworks alongside their modern-scientific parallels. Your child will learn to (a) name and use precise Sanskrit terms, (b) understand the observational basis for the classical framework, and (c) relate it to current scientific or scholarly understanding.

What you can do at home

Your child is learning a mental-math shortcut for ×11 problems. Help by giving them 2-digit numbers to multiply by 11 in spare moments (waiting for the bus, walking home). The aim is speed plus understanding — not memorisation.

Conversation prompts

Try one of these over a meal or walk:

1. "*What's one Sanskrit word you learned this week? What does it mean?*"
2. "*What did you find surprising in this module?*"
3. "*Is there anything you're not sure about? Let's look it up together.*"
4. "*How would you explain this to your grandmother / grandfather?*"

Questions parents often ask

Q: Is this religious instruction? A: No. The module presents Indian classical frameworks (cosmology, mathematics, astronomy, ecology) as historical knowledge systems with their own logic and method. The framing is scholarly and comparative, not devotional.

Q: Will my child be tested on Sanskrit? A: Sanskrit terms are introduced for precision, not for rote memorisation. The summative quiz asks students to *use* the terms — for example, "Which doṣa is dominant in the morning hours?" — rather than translate them.

Q: How does this connect to NCERT / board syllabus? A: This module supplements (does not replace) NCERT content. NEP 2020 §4.6 calls for the integration of Indian Knowledge Systems. The pack's `README.md` lists the specific NEP principles each module advances.

Reading further (for parents)

See `sources.md` in this pack for the full source list.

10-Day Lesson Plan

Lesson Plan — ×11 Multiplication — Ekādhikena Pūrveṇa (Senior)

10-day arc, 45 min per session.

Day 1 — The Hook — What Is the Trick?

Objective: Compute 23×11, 35×11 mentally using the split-and-sum method.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 2 — The Carry Case

Objective: Compute 87×11, 65×11 correctly using the carry rule.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 3 — Why It Works — The Algebra

Objective: Derive (10a+b)×11 = 100a + 10(a+b) + b and explain.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 4 — 3-Digit × 11

Objective: Compute 234×11 = 2_5_7_4 using the extended split method.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 5 — Speed Day

Objective: Complete a 50-problem mental-math sprint in 5 minutes.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 6 — Anurūpyena — ×12 to ×19

Objective: Extend the method to ×12 and ×13 using one-more-digit adjustments.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 7 — Limits of the Method

Objective: Recognise when traditional long multiplication is faster than the sūtra method.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 8 — Tirthaji and Historical Context

Objective: Explain who Tirthaji was and why scholars debate the "Vedic" attribution.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 9 — Mixed Practice

Objective: Solve a mixed problem set using best-fit method per problem.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 10 — Final Assessment

Objective: Take the summative quiz and reflect on which method works best.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Pacing notes

- Senior sessions are 45 min. Adjust if your block is different.
- Days 5 and 9 are buffer days — use to reinforce weaker concepts or extend stronger ones.
- The summative project (see `assessments/project-brief.md`) runs in parallel from Day 6 onward.

Differentiation

- **For advanced students:** see `teacher-notes.md` for extension prompts.
- **For students needing more support:** simplify activities, do more pair-work, allow oral instead of written responses.

Homework

Homework — ×11 Multiplication — Ekādhikena Pūrveṇa (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — The Hook — What Is the Trick?** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — The Carry Case** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Why It Works — The Algebra** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — 3-Digit × 11** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Speed Day** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Anurūpyena — ×12 to ×19** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Limits of the Method** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Tirthaji and Historical Context** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — Mixed Practice** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Final Assessment** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.